Towards Sensor-Aided Multi-View Reconstruction for High Accuracy Applications

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***Abstract*—We present the general idea of a computer vision structure-from-motion framework that makes use of sensor fusion to provide very accurate and efficient multi-view reconstruction results that can capture internal geometry. Given the increased ubiquity and cost-effectiveness of embedding sensors, such as positional sensors, into objects, it has become feasible to fuse such sensor data and camera-acquired data to vastly improve reconstruction quality and enable a number of novel applications for structure-from-motion. Application areas, which require very high accuracy, include medicine, robotics, security, and additive manufacturing (3D printing). Specific examples and initial results are discussed, followed by a discussion on proposed future work.**

***Keywords*—sensor fusion; embedded sensors; 3D Printing; computer vision; RGB-D; multi-view reconstruction; structure-from-motion; Kinect;**

I. Introduction

In the past few years, there has been a great increase in the amount of sensors that are embedded into every day devices on account of the positive trends in lower costs and miniaturization. For example, consider a modern Android® or iOS® phone whose internal sensors (GPS, camera, gyroscope, magnetometer, accelerometer, proximity, audio, and more) drastically overwhelm the bigger and less capable cellular phones of prior generations. This trend extends outside of industry and into research where other common sensors, including radar, sonar, LIDAR, infra-red, seismic, and magnetic have become utilized more often.

The ubiquity of such sensors and their data creates a *sensor fusion* problem. Sensor fusion involves combining data acquired from different sources in order to provide more accurate or complete information about the sensed target than if these sources were utilized individually. Fusion is non-trivial, and is a very relevant topic today in fields such as computer vision.

In computer vision, one specific instance of sensor fusion is the RGB-D camera, such as the *Microsoft Kinect*®, which jointly acquires color (RGB) data and depth (D) values for each pixel. While the Kinect has been a boon for research in reconstruction and many other fields, it has practical limitations for high-accuracy applications because depth estimates tend to be noisy, and without very accurate filtering, are generally not accurate enough to provide reliable data for up-and-coming applications in medicine, 3D printing and robotics. Furthermore, this approach suffers from a major problem that nearly all reconstruction techniques exhibit: the inability to capture internal structure because of occlusion. This can be a major drawback when reconstructing geometrically functional objects like ball bearings (see Figure 1).

Inspired by the gains of RGB-D fusion, we investigate the benefits of using positional sensors to assist in multi-view scene reconstruction. To that end, we present initial results on the development of a generalized framework for 3D scene reconstruction aided by any mix of positional data, such as RGB-D or sonar and photographic imagery. Furthermore, we explore the idea of placing these sensors internally in order to reconstruct internal structure. We show that fusing positional data with traditional images improves the accuracy of camera pose estimation and scene reconstruction, especially when dealing with texture-less or mono-colored objects, such as those produced by consumer level additive manufacturing processes (see Figure 1). This fusion also has the potential to capture internal structure as opposed to standard structure-from-motion approaches. Background and related work is discussed in II. Some concrete applications and results will be discussed in Section III. Conclusions and future work will be discussed in Section IV.



Fig. 1. A *Microsoft Kinect* analyzing various additively manufactured objects, which can be textureless and monocolored (presenting challenges for traditional reconstruction approaches). The ball bearing on the right exhibits functional geometry that is not always visible, and is unlikely to be reconstructed by modern techniques.

II. Background and Related Work

To our knowledge, we are the first to propose using internally embedded sensors for multi-view reconstruction. We provide a general background on computer vision and contributions towards scene reconstruction in Section II-A, and discuss recent work on fusing sensing techonology with imagery, specifically RGB-D cameras, in Section II-B. We will also discuss recent work on the imaging of internal geometry in Section II-C.

A. General Background on Scene Reconstruction

The broad field of computer vision includes important sub-fields such as object detection, tracking, and the multi-view reconstruction of scenes. The goal of multi-view scene reconstruction is to extract a 3D point cloud representing a scene when given multiple views (such as photographs) of the scene. Detailed analysis and comparisons between methods are available in the literature [1]. Most of these methods seek to create correspondences between views, usually by detecting features and tracking them from view to view. One of the main algorithms used to do this is Scale-Invariant Feature Transform (SIFT) [2]. For an excellent overview of many classical vision algorithms, the reader is referred to Hartley and Zisserman [3].

One drawback of current computer vision methods is that many are based on the mathematical optimization of initial parameter estimates to achieve accurate results. Though such optimization is provably necessary, such as in the case of the well known *bundle adjustment* [4] in structure-from-motion, the final accuracy is simply not enough for applications that require an extreme amount of accuracy. Furthermore, the density of these reconstructions often leaves something to be desired.

B. RGB-D Cameras

To alleviate the density problem, there has been interest in utilizing depth sensing technologies for object reconstruction for a long time [5], but it is only recently that the technology has become very affordable and easy to use with the release of the *Microsoft Kinect*® in late 2010. With it, came a plethora of reconstructions of people [6] and indoor environments [7]. One of the biggest successes is KinectFusion [8], which fuses depth data and RGB data from a movable Kinect in real time (using the GPU) to create a dense scene reconstruction as the user moves through the scene. Our work differs in two ways. First, we use structure-from-motion for increased accuracy in the face of noisy Kinect data. Second, our framework can fuse any source of positional information, including the Kinect, and by virtue potentially capture geometry that is not visible to the naked eye. In the next section, work on viewing such "hidden" geometry is discussed.

C. Internal Imaging

Research in internal imaging has been existant for a long time and has resulted in tremendous advances in both medicine and security. Breakthroughs in these domains have largely been a result of physics and biology research. For example, magnets are used to image the gastrointestinal track, radio waves produced by excited hydrogen atoms are used to image the brain (MRI), and x-rays (CAT scan) and small amounts of radioactive substances (PET scan) are used for tomography. All of these procedures were revolutionary and are now routine. Similar techniques and technologies have recently been used in tandem with computer vision to enhance security. For example, Taddei and Sequeira demonstrated that x-ray tomography equipment can be calibrated using automatic pose estimation (APE) on the silhouettes of shapes [9].

We are also motivated by structural health monitoring, which is concerned with using embedded sensor networks to evaluate structures such as buildings and bridges and provide feedback on cracks, torsion, and other instabilities. For a great overview see Tignola et al [10]. We believe the sensors and technology used in structural health monitoring will eventually be miniaturized and can thus be expanded upon to be utilized for geometric reconstruction on much smaller objects than bridges. This has motivated us to begin our preliminary exploration of using such internal sensor networks and fusing them with imagery to improve current reconstruction approaches.

III. Sensor-Aided Reconstruction

In order to provide a valid comparison point for our approach, the first thing we did was perform our own reconstructions using structure-from-motion, space carving, and KinectFusion using a new dataset. For these reconstructions, we used an additively manufactured chess piece and Utah teapot, both in red PLA plastic via the Makerbot Replicator 2®. Our motivation in creating this new dataset using 3D printed objects was *a)* we have ground-truth knowledge about the correct geometry *b)* we intend to show the potential benefits of embedding sensors as part of the manufacturing process and *c)* it allows for the object to be materialized by anyone who wishes to try their own physical approach (for example, KinectFusion).

Our results, shown in Figure 2, demonstrate the various problems with these standard approaches for the objects in (a) and (e). Structure-from-motion results, (b) and (f), are meshed reconstructions retrieved from running Visual SfM [11], [12] and using PMVS [13] to densify. While this approach does a decent job of accurately capturing some important details like the crown on the queen and the handle and spout of the teapot, it is clear that the reconstruction is full of holes and not dense enough. It is important to note that the lack of texture and color variance is one of the major problems for structure-from-motion since it largely depends on the presence of lots of unique features for tracking. Another common approach that doesn't depend on texture or color is space carving [14]. In images (c) and (g), you can see that although it does a great job of creating a dense, water-tight, reconstruction by virtue of the approach, it is not accurate enough to capture the sharp tips of the crown and none of the spout or handle of the teapot. Similarly, KinectFusion [8], (d) and (h), creates wonderfully dense objects but fails to capture small details due to the smallness of the objects, hardware limitations and noise.

In light of these results, we developed an alternative reconstruction pipeline which couples positional information with structure-from-motion. In general, accurate structure-from-motion based reconstruction typically relies on accurate

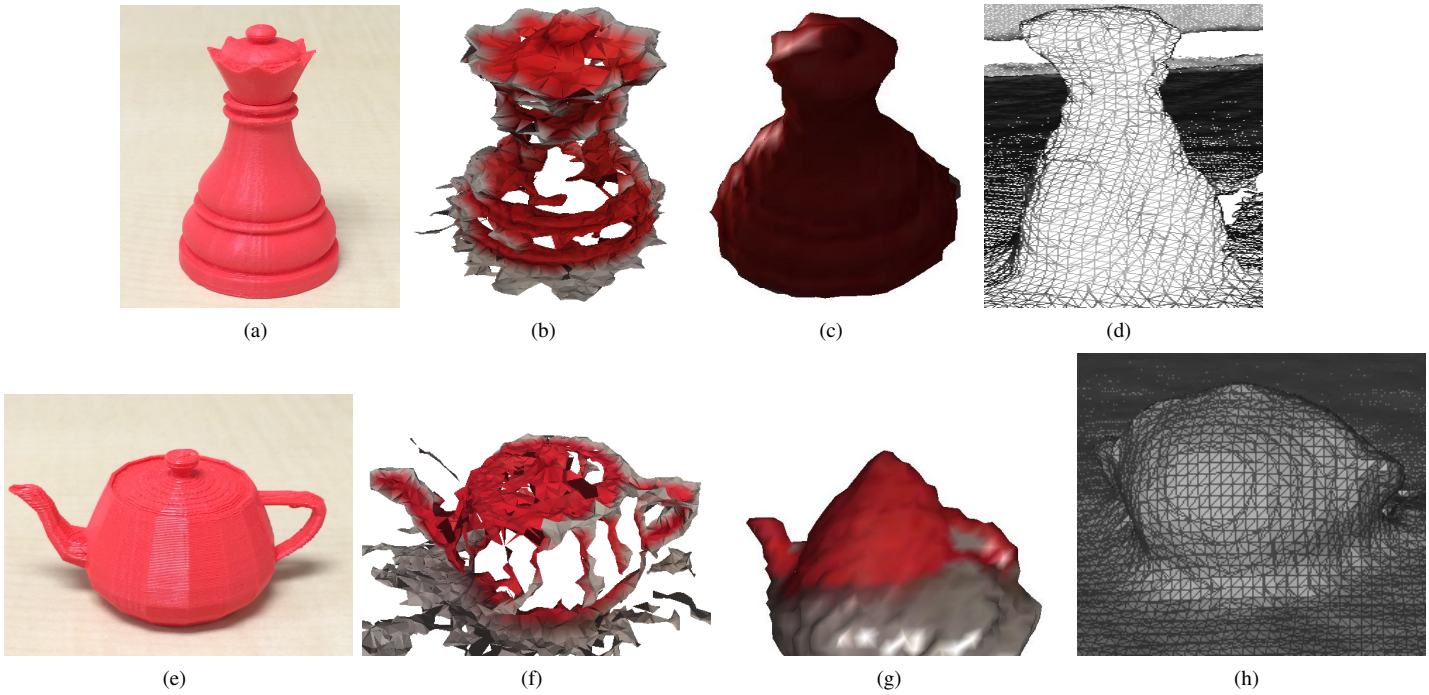


Fig. 2. Input (a), structure-from-motion (b), space carving (c), and KinectFusion (d) reconstructions for a 3D-printed red chess piece. The same is shown for the "Utah teapot" in (e) - (h).

feature tracking [3]. A feature track is a set of pixel positions representing a scene point tracked over a set of images. Given a 3D position computed from multi-view stereo, its reprojection error with respect to its corresponding feature track is the only valid metric to assess error, in the absence of ground truth. Highly inaccurate individual track positions adversely affect subsequent camera pose estimation and structure computation, as well as bundle adjustment. Such inaccuracies can be improved upon by including external sensor information, such as positional information, into solving for scene reconstruction. The advantage of counting with embedded positional information inside an object is that it avoids having to compute accurate feature tracks in order to perform camera parameter and structure computation.

A diagram of our pipeline is shown in Fig 3. The process begins by collecting both the positional sensor data and image data. Provided with a mechanism for locating the positional sensor in each image, the accurate position information is used to perform camera pose estimation. This leads to accurate camera rotation and translation measurements and is void of the inaccuracies present when using feature tracks to estimate camera pose. Feature tracking is performed on the image data and is combined with the accurate camera data to perform triangulation of the scenes 3D structure. Errors in the feature tracking stage are manifested as inaccurate scene points and bundle adjustment is used to optimize the reprojection error of the given structure and camera parameters. After bundle adjustment has been performed, the scene geometry can be stored and manipulated using standard modeling techniques.

Initial results show the positive effect of embedded positional sensors on reconstruction. Figure 4 shows a synthetic dataset consisting of internal position sensors (yellow) inside a cube viewed by a randomly-placed set of cameras (green). Surface sensors on the cube are shown in red. Using a pipeline

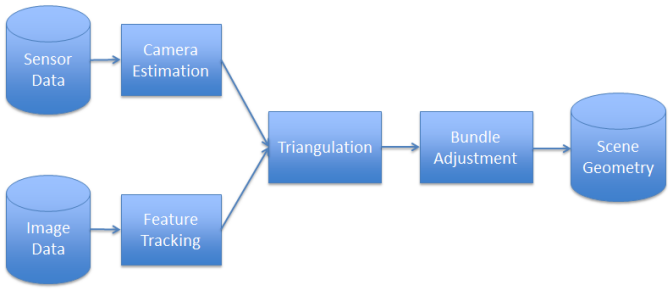


Fig. 3. Our proposed general scene reconstruction pipeline including positional information. Given known and accurate position information, it is first used to perform camera pose estimation, in order to compute accurate camera rotation and translation measurements. Then, feature tracking is performed on the image data and is combined with the accurate camera data to perform triangulation of the scenes 3D structure. Finally, bundle adjustment is used to optimize the reprojection error of the given structure and camera parameters.

which computes camera parameters from this information, we were able to achieve almost a zero reprojection error.

IV. Conclusion and Future Work

This paper presented the general idea of using sensor fusion as a strong tool for improving accuracy in computer vision structure-from-motion with the end goal of enabling high accuracy applications. Concrete results were shown for synthetic data, where a simulated object with internal and external position sensors was used to very accurately estimate the set of cameras viewing the object. Given these initial results, we believe the future is bright with regards to fusing sensor measurements for improved multi-view reconstruction, which is the focus of our ongoing work.

We have identified a number of uses in potential

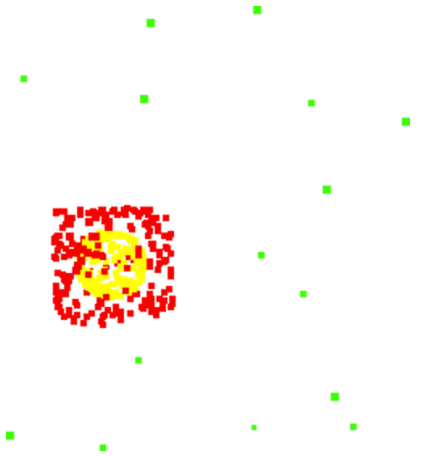


Fig. 4. This figure demonstrates that accurate internal sensors can improve a scene reconstruction. Synthetic internal positional sensors (shown in yellow) are used to generate accurate camera data (camera positions are shown in green). The external red box structure was computed using standard structure from motion techniques. The sensor fusion allows for very accurate reconstructions. The average reprojection error for this synthetic scene was $6.66145e^{-5}$.

applications. One is the additive manufacturing process. By embedding positional sensors as part of the 3D-printing process, a whole host of opportunities open up. First, you can monitor and analyze the object during the printing process and verify key geometric qualities, such as distances or angles. Secondly, if the sensors are miniaturized to a sufficient degree and placed very densely, it becomes unnecessary to even use structure-from-motion or other techniques since a meshed point cloud of sensor locations can be used as a reconstruction by itself. Third, a designer could manipulate the printed object with real world tools, such as chisels and saws, and be able to "scan" the object back into virtual space. A similar process already occurs in structural health monitoring where sensors are mixed with concrete; it is our belief that is only a matter of time before the technology is miniaturized enough for small scale objects and additive manufacturing.

Furthermore, at that miniaturized scale, we can expand the concept of structural health monitoring to medical applications and devices. By embedding sensors in artificial human parts, such as hearts and prosthetics, we enable the medical community to non-invasively monitor any defects that may occur by periodically reconstructing the object and analyzing it.

We strongly believe that miniaturized positional sensors are achievable with some combination of modern technologies such as ultrasound, magnets, and piezoelectrics. Our future work will focus on using prototyped positional sensors to proof-of-concept our approach and its revolutionary applications.

ACKNOWLEDGMENT

This work was performed with support from Sandia National Laboratory. Special thanks to colleagues at the Institute for Data Analysis and Visualization, for the useful discussions and support.

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